

Evaporation

Final Lab Report

Unit Operations Lab 1

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Group 2

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1. Executive Summary

The objective of this experiment was to determine the performance characteristics of a single-effect evaporator by evaluating the influence of vacuum pressure on capacity and steam economy. Since evaporation is an energy-intensive unit operation, understanding these relationships is critical for designing and operating efficient evaporation systems in industrial settings.

Experimental trials were conducted using 10 wt% sugar solution at three vacuum pressures: 20, 22, and 25 inHg, corresponding to boiling temperatures of 71, 65, and 63 °C, respectively. The evaporator capacity, defined as the rate of water vaporized, increased with decreasing operating pressure, rising from 2.04 kg h⁻¹ at 20 inHg to 3.48 kg h⁻¹ at 25 inHg. Similarly, the steam economy, expressed as the ratio of water vaporized to steam consumed, improved from 0.447 to 0.725 across the same range. These results confirm that operating at higher vacuum levels enhances the driving force for heat transfer, leading to higher production rates and better energy utilization.

We can conclude that reducing the operating pressure significantly improves evaporator performance. The observed increase in both capacity and economy aligns with theoretical expectations, although the economy did not exceed unity; this is consistent with the constraints of single-effect evaporation. Overall, the results emphasize that vacuum optimization is a key strategy for improving energy efficiency and productivity in evaporation processes. They also highlight the potential benefits of adopting multiple-effect evaporators in an industrial setting.

2. Introduction

Experimental objective

This experiment aims to have a clear picture of how a single-effect evaporator functions and to test its efficiency under various operating conditions. In particular, the experiment aims at investigating the influence of vacuum pressure on the boiling point of the solution, the speed of vaporization, and the effectiveness of the evaporator in general. Product collected, temperature changes at varying vacuum levels, and measuring the steam consumed, we will be able to calculate the important performance indicators, including capacity (quantity of water vaporized divided by time) and economy (quantity of water vaporized divided by quantity of steam consumed). Another point of the experiment is the use of energy and mass balances that are fundamental in chemical engineering to analyze and scale processes.

Significance of the experiment

This experiment is significant in the sense that the evaporation process constitutes one of the most frequent unit operations in chemical engineering. It provides a practical situation concerning the way heat transfer and phase change work together in an industrial setting. With the help of this lab, we become acquainted with what occurs to the boiling point as well as to the overall consequential pressure. Correlating the chosen theoretical concepts, e.g., energy balances, is also possible in the experiment. The experiment makes us realize why it was problematic when evaporation had to be scaled up from lab scale to plant scale.

Brief description of apparatus and experiment

The methods used in the experiment are a single-effect evaporator, which is utilized in concentrating on a 10 wt% sugar solution. The equipment comprises a separator in which the solution boils, a condenser to harvest the water vapor, a steam heater to supply the energy required, and lastly, a vacuum pump to work on the temperature of boiling to lower the pressure.

Those points in the system where temperature changes are to be checked are occupied by thermocouples.

In the experiment, steam is introduced into the system to make the solution hot, and the vacuum pump is employed to raise and lower the pressure in the separator. When the solution is boiled, the steam that is formed gets stored in a condenser. The experiment would be conducted in three vacuum pressures, and this would enable us to observe that the pressure is reduced the higher the temperature variation between the heating steam and the boiling solution, thus enhancing the pace of heat transfer. We find the mass of water vapor condensed and the mass of steam condensed at every vacuum setting. The data is then applied to the calculation of the capacity and economy of the evaporator, as well as the analysis of performance depending on the circumstances under which it is operated.

Industrial Applications

Evaporation is a process that has quite a lot of industrial applications. It is applied to liquids like fruit juices, milk, and syrups within the food industry to reduce the volume and lengthen their shelf life. Evaporators in the pharmaceutical industry have been used in the purification and concentration stages, like in making active ingredients or extracting solvents. They find application in the chemical industry as a way of separating volatile solvents and nonvolatile products or converting chemical solutions into a concentrated form to proceed with additional processing. Evaporation plays a big role in treating and desalinating water, which serves to take away salt in seawater or to concentrate wastewater as one of the ways to ensure easier disposal or re-industrial use of it.

We can learn the main principles of working on the large-scale implementation by examining how one of the lab-scale evaporators can be used. The experience associated with this experiment brings up insights on how engineers enhance efficiency in the processes, reduce energy consumption, and create cost-efficient operations in actual industries.

3. Theory

Evaporation concentrates a non-volatile solute, in this instance, sucrose, by boiling off the volatile solvent, water. In a steam-heated, single-effect evaporator, saturated steam condenses on one side of the heating surface while the liquid on the other side boils. Operating under a vacuum lowers the boiling temperature, increasing the temperature driving force and thus the heat-transfer rate. Performance is characterized by capacity in units of kg water vaporized per hour, as well as steam economy, in units of distillate produced per kg steam consumed. In this experiment, we are determining how vacuum and absolute pressure affect capacity and economy for a sucrose solution in a single-effect unit.

The governing relationships in this experiment include the heat transfer coefficient and the energy balance equation. The performance is governed by the capacity and economy equations.

<i>Table 1. Equations</i>		
Heat Transfer	$q = UA \Delta T$	Where U is the overall heat-transfer coefficient, A is the heat-transfer area, and ΔT is the temperature difference.
Energy Balance	$q = \dot{m}_v \lambda_w + \dot{m}_f c_p (T_b - T_f)$	Where $\dot{m}_v$ is the vapor (distillate) mass

		flow, λ_w is the latent heat of vaporization of water at an absolute pressure, T_b is the boiling temperature in the separator, and T_f is the feed temperature.
Capacity	$\dot{m}_v$	Where $\dot{m}_v$ is the vapor (distillate) mass flow.
Stream Economy	$\frac{\dot{m}_v}{\dot{m}_s}$	Where $\dot{m}_s$ is the steam condensed.

To convert the experimental measurement into results, the capacity needs to be calculated by using the distillate mass flow by collecting the condenser product for a timed interval. Converting the measured volume to mass using the water density at that measured temperature. The capacity will need to be plotted versus absolute pressure in the vacuum. As absolute pressure decreases, a larger effective driving force is thus created. Therefore, the temperature difference yields a higher capacity.

Steam consumption using cooling water calorimetry will be used to calculate how much steam is condensed on the heating side. This will be key in calculating the economy, as the higher the mass flow of condensed steam, the lower the economy.

By graphing the capacity vs. the absolute pressure and the economy vs. the absolute pressure, as well as calculating the energy balance of the measured flows and temperature, we can accurately display the results of this experiment.

4. Results

Three steady state runs were collected at vacuum pressures of 20, 22, and 25 inHg. These pressures lead to temperatures of 71, 65, and 63 °C, respectively. The primary outcomes are that the capacity and the steam economy both improved as the vacuum was increased. The capacity specifically rose from 2.04 kg h⁻¹ at 20 inHg to 3.48 kg h⁻¹ at 25 inHg, while the economy increased from 0.447 to 0.725, indicating substantially less steam was required per unit of evaporation at a deeper vacuum, consistent with a larger effective temperature driving force and reduced boiling temperature.

Based on the experimental data, the vacuum pressure increased, the measured boiling temperature and capacity fell, consistent with a reduced temperature driving force at a higher absolute pressure. This led to the capacity vs. vacuum and economy vs. vacuum to be approximately linear over the tested range.

5. Discussion

The results of this experiment demonstrated clear trends that are consistent with the theoretical performance of single-effect evaporators. As vacuum pressure increased from 20 to 25 inHg, the boiling temperature of the sugar solution decreased from 71 °C to 63 °C. The reduction in boiling temperature increases the temperature difference between the heating system and the boiling liquid, enhancing the driving force for heat transfer. The heat capacity also increased from 2.04 to 3.48 kg h⁻¹, while the economy rose from 0.447 to 0.725. Both magnitudes and trends agree with the principles of evaporation and heat transfer. These trends emphasize that lowering operating pressure leads to higher heat transfer rates and improved evaporator performance.

Although the observed results matched theoretical expectations, the steam economy remained below unity, reflecting the inefficiency of single-effect evaporation. For industrial-scale multiple-effect evaporators, economy is well over one, which highlights the energy savings achieved by reusing vapor across stages. The difference between our results and other reported values can be attributed to the single-effect design of the laboratory apparatus rather than experimental error.

Possible outcomes of experimental error include inaccuracies in measuring condensate volumes, incomplete recovery of condensate, and possible uncertainty in pressure gauge or thermometer readings. If any leaks were present in the vacuum system, this could also affect the steady state operating pressure, therefore affecting the boiling point and heat transfer rate. While these potential errors could have affected our data, the trends in our reported values are consistent, and this suggests that the conclusions drawn remain valid. Improvements could include more precise flow measurement devices and enhanced insulation to minimize heat losses and securing the vacuum knob to avoid turning while the apparatus operates.

The assumption of constant sugar concentration, maintained by adding deionized water as makeup, was reliable for isolating the effects of vacuum pressure. However, this assumption would not be valid in an industry setting, where feed concentration and viscosity would also influence capacity and economy. Future studies could investigate these factors to provide a more realistic industry profile.

Overall, the results confirm theoretical expectations, lowering operating pressure enhances both capacity and steam economy. It can also be seen that single-effect designs prevent achieving economies greater than unity. These findings reinforce the importance of multiple-effect designs in practice and suggest that further work should be done to compare laboratory-scale results to predictive models that incorporate non-ideal solution behavior.

Post-lab Questions:

1. Why is pure deionized water used as the makeup solution?

Deionized water is added to get pure water with an equal level of sugar concentration in the separator throughout the experiment. This maintains a stable system, and only vacuum

pressure changes. The boiling point would be higher, and it would be difficult to analyze the concentration, permitting it to increase. In industry, it is a natural increase of concentration because it is the aim to manufacture a more viscous product.

2. Does the heat of vaporization of steam depend on pressure? Where is it found on the steam tables?

Yes, it is true here that the size of the vaporization heat reduces with increasing pressure. Increased pressure results in increased boiling temperature and reduced energy to vaporize every kilogram of water. The thermal difference between the enthalpy of saturated vapor and saturated liquid is found in the saturated properties of steam tables as the difference between the saturated properties of vaporization and the enthalpy of saturated liquid, reported as h_{fg}

3. What other factors besides vacuum setting affect evaporator performance?

The capacity and economy of the evaporator are influenced by factors other than vacuum setting, which include the difference in concentrations of the sugars, as the higher the concentration, the higher the boiling point and the lower the temperature driving force. The rate of steam flow is also necessary, since it figures in the quantity of heat provided; more may be put in at the cost of less economy in case of inefficient consumption. Moreover, there is a contribution of the liquid level in the separator to the hydrostatic head that partially increases the boiling temperature. Lastly, heat transfer may be impaired because of fouling on the heating surface or alteration in cooling water flow, and this may adversely impact performance.

6. Appendix

a. Appendix I: Data Tabulation/ Graphs

Table 2. Experimental Data

Run	Pressure (inHg)	Pressure (mmHg)	P _{abs} (mmHg)	P _{abs} (kPa)	T _b (°C)	Time (s)	Condensate (mL)
1	25	635.0	125.0	16.67	63	300	290
2	22	558.8	201.2	26.82	65	310	240
3	20	508.0	252.0	33.60	71	300	170

Table 3. Data Summary

Run	Capacity (kg/h)	Economy	$\dot{m}_s$ (kh/h)	Concentrate draw (mL)
1	3.48	0.725	4.80	400
2	2.782	0.558	4.99	430
3	2.04	0.447	4.56	380

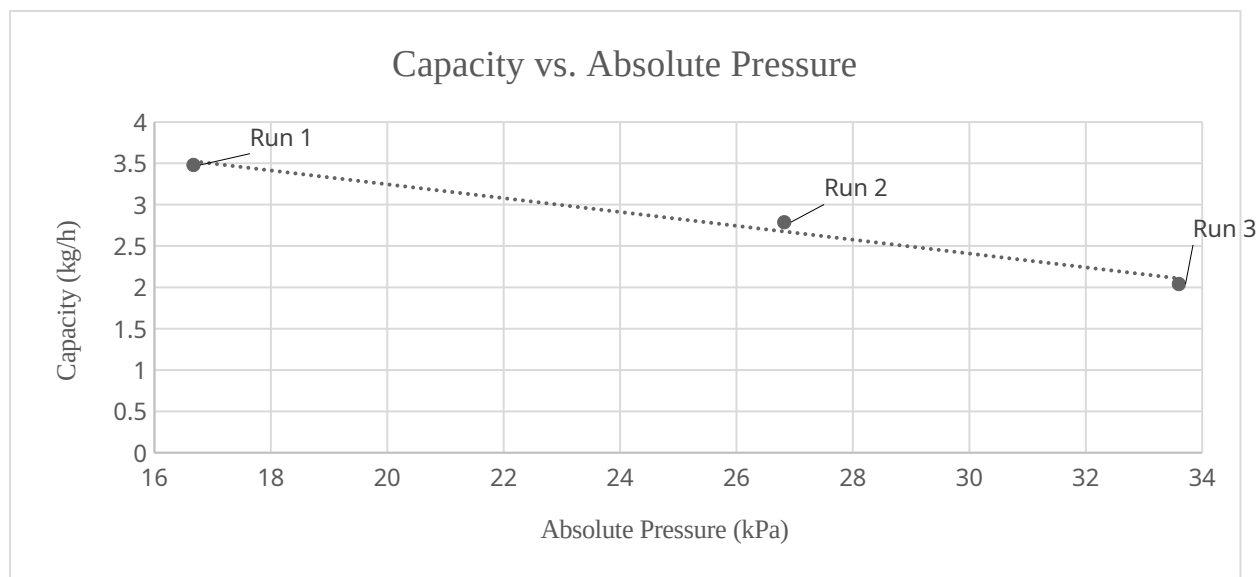
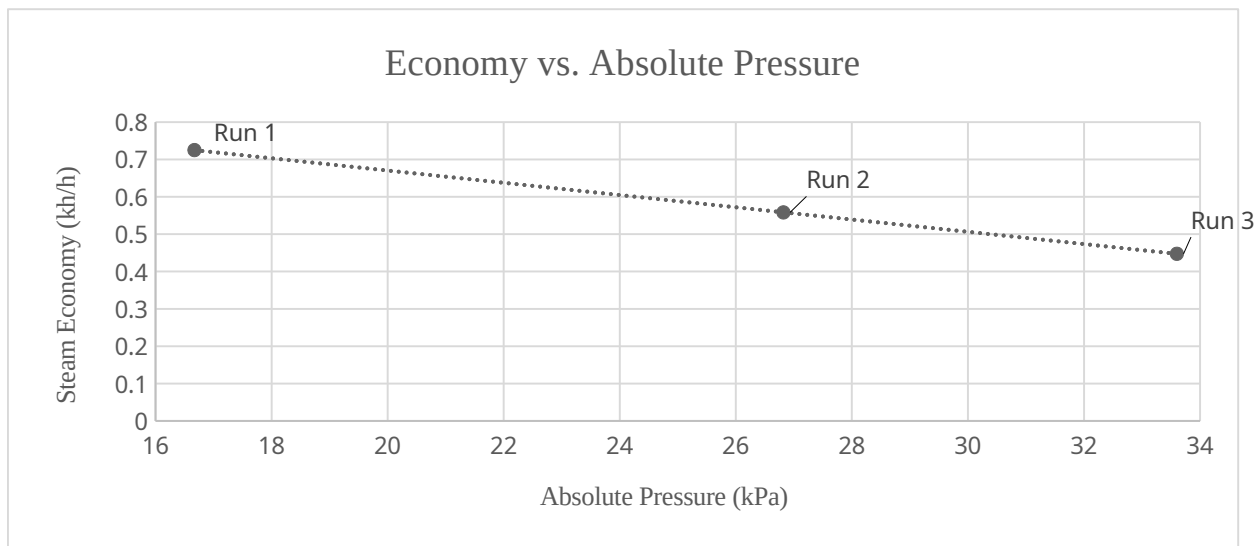
Figure 1: Capacity vs. Absolute Pressure graph showing linearity across the three runs and the negative trendline

Figure 2: Economy vs. Absolute Pressure graph showing the linearity across the three runs and the negative trendline.



b. Appendix II: Error Analysis

Capacity uncertainty is dominated by the volume and timing. Since the volume and the time of the run is not uniform across all three runs, that can cause error in the readings and therefore errors in the overall capacity values. Economy uncertainty is additionally affected by the steam-side calorimetry, as the cooling-water flow and the change of temperature are unspecified, which can cause the reported economy values to be incorrect.

c. Appendix III: Sample Calculations

1. Convert vacuum to absolute pressure

$$\text{Vacuum (mmHg)} = 24 \text{ inHg} \times 25.4 \frac{\text{mm}}{\text{in}} = 635.0 \text{ mmHg} \quad P_{\text{abs (mmHg)}} = 760 - 635.0 = 125.0 \text{ mmHg}$$

$$P_{\text{abs (kPa)}} = 125.0 \times 0.133322 = 16.67 \text{ kPa} \quad P_{\text{abs (atm)}} = \frac{125.0}{760} = 0.164 \text{ atm}$$

2. Capacity

$$\text{Volume} \rightarrow \text{mass} : m_d = \rho V_d = 0.997 \frac{\text{kg}}{\text{L}} \times 0.290 \text{ L} = 0.28913 \text{ kg}$$

$$\text{Seconds} \rightarrow \text{hours} : t_h = 300 \text{ s} \times \frac{1 \text{ h}}{3600 \text{ s}} = 0.08333 \text{ h}$$

$$\text{Capacity} : \dot{m}_v = \frac{\dot{m}_d}{t_h} = \frac{0.28913}{0.08333} = 3.47 \frac{\text{kg}}{\text{h}}$$

3. Steam Economy

$$\text{Economy} = \frac{\dot{m}_v}{\dot{m}_s} = \frac{3.48}{4.80} = 0.725$$

7. References

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